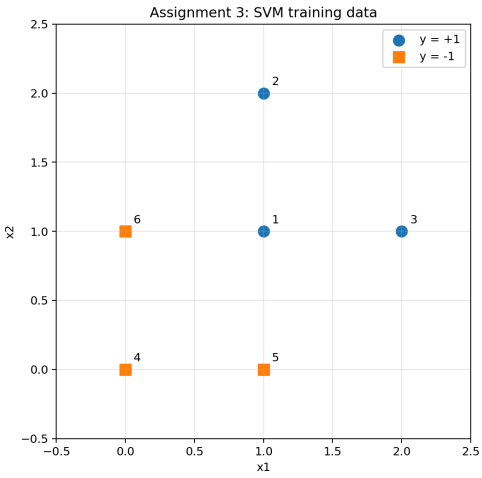
Q1



(a)

- Point 1, 5, and 6 are the support vectors because they are the points from each class closest to each other class

(b)

- W * x + b = 0
- (1,1): w1 + w2 + b = 1 -> a
- (0,1): w2 + b = -1 -> b
- (1,0): w1 + b = -1 -> c
- b - c: w2 = w1
- b - a: w1 = 2
- w = (2,2)
- b: 2+b = -1
- b = -3

Margin

||w|| = (2^2 + 2^2)^½ = 2(2)^½
= 1 / ||w|| = 1 / (2(2)^½)) = (2)^½ / 4 = 0.354

Q2

(a)

- Equal distance ties are solved by getting the first occurrence using argmin
- An empty cluster keeps the same centroid, in code this is done by copying the input centroid data and the centroid only gets updated if there is a cluster greater than 0.

(b)

- As k increases the total square distance from each point to the centroid decreases which means we have tighter clusters as k is increasing.

Q3

Two method accuracy

- The average accuracy for the baseline is 0.82 and for k-nn repaired is 0.86. This is only a small increase of 3.6 percent. Overall I think this means that the repair didn’t make the labels more correct. However, it did make them consistent within each neighborhood.